

Supplementary Movies and Supplementary Text for “Shape Transformations of Vesicles induced by Swim Pressure”

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CAPTIONS FOR MOVIES S1 TO S4

Supplementary Movie 1 This movie shows the active pearling behavior corresponding to Fig. 4b of the main text. The self-propulsion velocity is $v_0 = 17.5 \mu\text{m/s}$. The other parameters are at their default values given in Tab. S1. The particles tend to move in clusters. The movie shows how the clusters emerge, vanish, and move along the vesicle, splitting into smaller clusters and merging into larger ones; this behavior is more clearly observed in Supplementary Movie 2, which is the same except that only the particles are shown. The active vesicle pearling is even more pronounced for larger systems, see Supplementary Movies 3 and 4.

Supplementary Movie 2 Same movie as Supplementary Movie 2, but showing the particles only. The different clusters are color coded; two particles that are within a distance $1.2\sigma_{\text{pp}}$ are considered to belong to the same cluster, and grey particles do not belong to any cluster.

Supplementary Movie 3 Active pearling behavior for a larger vesicle. The number of particles is $N = 200$, and the membrane area is $A_0 = 4\pi R_0^2$ with $R_0 = 4/3 \mu\text{m}$. It is seen that the active pearling behavior is much more pronounced. Particles move almost exclusively in clusters, which tend to travel in an oscillatory yet stochastic fashion from pole to pole. Supplementary Movie 3 has the clusters color coded, to elucidate their dynamics. Other parameters: $v_0 = 100 \mu\text{m/s}$, $D_t = 1 \mu\text{m}^2/\text{s}$. The number of vertices of the membrane mesh is $N_v = 2400$. The remaining parameters are the same as in Tab. S1.

Supplementary Movie 4 Same movie as Supplementary Movie 2, but showing the particles only. The different clusters are color coded; two particles that are within a distance $1.2\sigma_{\text{pp}}$ are considered to belong to the same cluster, and grey particles do not belong to any cluster. Clearly, particles tend to move within a cluster. These clusters are predominantly nucleated from the clusters at the poles.

MODEL AND METHODS

Active particles The position $\mathbf{r}_i$ and orientation $\mathbf{e}_i$ of each active Brownian particle i follows the overdamped

translational and rotational Langevin equations:

$$\dot{\mathbf{r}}_i = \mathbf{F}_p^{(i)}(\{\mathbf{r}_i\}) \frac{D_t}{k_B T} + \sqrt{2D_t} \boldsymbol{\chi}_t(t) + v_0 \mathbf{e}_i, \quad (\text{S1})$$

$$\dot{\mathbf{e}}_i = \sqrt{2D_r} \boldsymbol{\chi}_r(t) \times \mathbf{e}_i. \quad (\text{S2})$$

Here, $\mathbf{F}_p^{(i)}(\{\mathbf{r}_i\})$ is the total force on particle i , arising both from the interactions with the other particles and from the interaction with the membrane, v_0 is the self-propulsion velocity, $k_B T = \beta^{-1}$ is Boltzmann's constant times temperature, D_t and D_r are the translational and rotational diffusion constants of the particles, and $\boldsymbol{\chi}_t$ and $\boldsymbol{\chi}_r$ denote the Gaussian white noise, whose Cartesian components χ_i obey $\langle \chi_i(t), \chi_j(t') \rangle = \delta_{ij} \delta(t - t')$.

The force on particle i is $\mathbf{F}_p^{(i)}(\{\mathbf{r}_i\}) = -\nabla_i U_p(\{\mathbf{r}_i\})$, where the total energy $U_p(\{\mathbf{r}_i\})$ of the particles is the sum of the particle-particle interaction energy $U_{\text{pp}}(\{\mathbf{r}_i\})$ and the particle-vertex interaction energy $U_{\text{pv}}(\{\mathbf{r}_i, \mathbf{q}_i\})$, which depends on the positions $\{\mathbf{r}_i\}$ of the active particles and the positions $\{\mathbf{q}_i\}$ of the vertices of the mesh that model the membrane (see below). The particle-particle interaction is given by

$$U_{\text{pp}}(\{\mathbf{r}_i\}) = \frac{1}{2} k_{\text{pp}} \sum_{i < j} (r_{ij} - \sigma_{\text{pp}})^2 \theta(\sigma_{\text{pp}} - r_{ij}), \quad (\text{S3})$$

where r_{ij} is the distance between particles i and j , and the stiffness k_{pp} governs the steepness of the repulsive interaction for distances smaller than the effective particle diameter σ_{pp} ; $\theta(x)$ is the Heaviside function. For the particle-vertex interaction we use the same form of the potential:

$$U_{\text{pv}}(\{\mathbf{r}_i\}, \{\mathbf{q}_i\}) = \frac{1}{2} k_{\text{pv}} \sum_{i,j} (r_{ij} - \sigma_{\text{pv}})^2 \theta(\sigma_{\text{pv}} - r_{ij}), \quad (\text{S4})$$

where $r_{ij} = |\mathbf{r}_i - \mathbf{q}_j|$ is the distance between the position of particle i and that of vertex j of the membrane, and σ_{pv} is the cross-section of the interaction between a particle and a vertex. For simplicity, we have chosen $k_{\text{pp}} = k_{\text{pv}}$ and $\sigma_{\text{pp}} = \sigma_{\text{pv}}$ (see Table S1 for all parameter values).

Vesicle The total free energy of the vesicle is given by

$$\mathcal{H} = \mathcal{H}_{\text{bend}} + \mathcal{H}_A + \mathcal{H}_V. \quad (\text{S5})$$

Here, $\mathcal{H}_{\text{bend}}$ is the Helfrich bending free energy [1]

$$\mathcal{H}_{\text{bend}} = 2\kappa \oint H^2 dS, \quad (\text{S6})$$

where κ is the bending rigidity of the membrane, H the mean curvature of the membrane, and the integral has to be performed over the whole surface. The other terms are given by

$$\mathcal{H}_A = \frac{1}{2} M_A ((A - A_0)/A_0)^2, \quad (\text{S7})$$

$$\mathcal{H}_V = \frac{1}{2} M_V \left((\hat{V} - \hat{V}_0)/\hat{V}_0 \right)^2. \quad (\text{S8})$$

These terms make it possible to constrain the membrane and the volume of the vesicle, respectively. The reduced volume $\hat{V} \equiv V/V^*$ is the volume of the vesicle V divided by the volume V^* of a sphere with the same area A as that of the vesicle.

To simulate the membrane shape dynamics, we employ the dynamic triangulation method of Gompper *et al.* [2]. In this method, the membrane is discretised into a triangular mesh. Links between two vertices are described via a vertex-vertex interaction potential $U_{vv}(\{\mathbf{q}_i\})$ [3] and, to model the fluidity of the membrane, links can be flipped between two possible diagonals of two adjacent triangles via a Metropolis Monte Carlo scheme [2]. In between the bond flips, the vertices obey the overdamped Langevin equation:

$$\dot{\mathbf{q}}_i = \mathbf{F}_v^{(i)}(\{\mathbf{q}_i\}) \frac{D_m}{k_B T} + \sqrt{2D_m} \chi_v(t) \quad (\text{S9})$$

where $\mathbf{F}_v^{(i)} = -\nabla_i U_v(\{\mathbf{q}_i\})$ is the total force acting on vertex v , D_m is the diffusion constant of the vertices, and χ_m denotes Gaussian white noise, with its Cartesian components χ_i satisfying $\langle \chi_i(t), \chi_j(t') \rangle = \delta_{ij} \delta(t - t')$.

The total force acting on vertex i is given by

$$\mathbf{F}_v^{(i)}(\{\mathbf{q}_i\}) = \mathbf{F}_{vv}^{(i)}(\{\mathbf{q}_i\}) + \mathbf{F}_{pv}^{(i)}(\{\mathbf{r}_i, \{\mathbf{q}_i\}\}) + \mathbf{F}_A^{(i)}(\{\mathbf{q}_i\}) + \mathbf{F}_V^{(i)}(\{\mathbf{q}_i\}) + \mathbf{F}_{\text{bend}}^{(i)}(\{\mathbf{q}_i\}). \quad (\text{S10})$$

Here, $\mathbf{F}_{vv}^{(i)}(\{\mathbf{q}_i\}) = -\nabla_i U_{vv}(\{\mathbf{q}_i\})$ is the force on vertex i arising from the vertex-vertex interactions, $\mathbf{F}_{pv}^{(i)}(\{\mathbf{r}_i, \{\mathbf{q}_i\}\}) = -\nabla_{\mathbf{q}_i} U_{pv}(\{\mathbf{r}_i, \{\mathbf{q}_i\}\})$ is the force arising from the interactions with the particles with $U_{pv}(\{\mathbf{r}_i, \{\mathbf{q}_i\}\})$ given by Eq. S4, $\mathbf{F}_A^{(i)}(\{\mathbf{q}_i\}) = -\nabla_i \mathcal{H}_A(\{\mathbf{q}_i\})$ is the force arising from the area constraint, and $\mathbf{F}_V^{(i)}(\{\mathbf{q}_i\}) = -\nabla_i \mathcal{H}_V(\{\mathbf{q}_i\})$ is the force arising from the volume constraint. Lastly, $\mathbf{F}_{\text{bend}}(\{\mathbf{q}_i\})$ is the bending force, which is the bending pressure $\Delta p_{\text{bend}}(\{\mathbf{q}_i\})$ times the Voronoi area associated with vertex i . The bending pressure is obtained from the first variation of the Helfrich free energy, Eq. S6 [4]:

$$\Delta p_{\text{bend}}(\{\mathbf{q}_i\}) = \kappa [4H(H^2 - G) + 2\nabla^2 H], \quad (\text{S11})$$

where G is the Gaussian curvature. Here, both H and G depend on $\{\mathbf{q}_i\}$, which has been dropped in the above equation for notational clarity.

In the simulations, Eqs. S1, S2 and S9 are propagated via a simple Euler scheme using a time step Δt :

$$\Delta \mathbf{r}_i = \mathbf{F}_p^{(i)}(\{\mathbf{r}_i\}) \frac{D_t}{k_B T} \Delta t + \sqrt{2D_t \Delta t} \chi_t(t) + v_0 \mathbf{e}_i \Delta t, \quad (\text{S12})$$

$$\Delta \mathbf{e}_i = \sqrt{2D_r \Delta t} \chi_r(t) \times \mathbf{e}_i, \quad (\text{S13})$$

$$\Delta \mathbf{q}_i = \mathbf{F}_v^{(i)}(\{\mathbf{q}_i\}) \frac{D_m}{k_B T} \Delta t + \sqrt{2D_m \Delta t} \chi_m(t). \quad (\text{S14})$$

The values of the parameters used in the simulations are listed in Table S1.

PROPULSION FORCE

Each active particle is propelled by a propulsion force f_{prop} , which in the absence of any other force impinging on the particle yields a self-propulsion velocity v_0 . At this steady-state velocity, the propulsion force is counterbalanced by the friction force $f_{\text{fric}} = -\gamma v_0$, where γ is the friction coefficient which via the fluctuation-dissipation theorem is given by $\gamma = k_B T / D_t$, where $k_B T$ is Boltzmann's constant k_B times temperature T and D_t is the translational diffusion constant. This yields $f_{\text{prop}} = -f_{\text{fric}} = \gamma v_0 = k_B T / D_t$, as used in the main text.

SHAPE ORDER PARAMETERS

We used a combination of order parameters to classify the vesicle shapes into prolate, oblate and stomatocyte.

The classification is based on the shape tensor $Q_{\alpha\beta}$, which is defined by the positions of the N vertices of the membrane mesh:

$$Q_{\alpha\beta} = \frac{1}{N} \sum_{i=1}^N (\mathbf{r}_{i\alpha} - \mathbf{r}_\alpha^{\text{cm}}) (\mathbf{r}_{i\beta} - \mathbf{r}_\beta^{\text{cm}}), \quad (\text{S15})$$

where $\mathbf{r}_{i\alpha}$ is the Cartesian component $\alpha = x, y, z$ of the position vector of vertex i and $\mathbf{r}_\alpha^{\text{cm}}$ is the Cartesian component α of the position of the center mass of all the N vertices. We denote the three eigenvalues of the tensor $Q_{\alpha\beta}$ by λ_1, λ_2 , and λ_3 , defined such that $\lambda_1 \leq \lambda_2 \leq \lambda_3$.

These eigenvalues are used to define two order parameters, which are widely used in polymer studies [7, 8]: the prolateness $\mathbb{P}$ and the asphericity $\mathbb{A}$.

The prolateness $-1 \leq \mathbb{P} \leq 1$ makes it possible to distinguish between prolate and oblate shapes. It is defined as

$$\mathbb{P} = \frac{(2\lambda_1 - \lambda_2 - \lambda_3)(2\lambda_2 - \lambda_1 - \lambda_3)(2\lambda_3 - \lambda_1 - \lambda_2)}{2(\lambda_1^2 + \lambda_2^2 + \lambda_3^2 - \lambda_1\lambda_2 - \lambda_1\lambda_3 - \lambda_2\lambda_3)^{3/2}}. \quad (\text{S16})$$

For prolate shapes $\mathbb{P} > 0$ while for oblate shapes $\mathbb{P} < 0$.

Parameter	Description	Value
Active particles		
D_t	Translational diffusion constant	$0.1\mu\text{m}^2/\text{s}$
D_r	Rotational diffusion constant	$1/\text{s}$
k_{pp}	Stiffness interaction potential	$10^6 k_B T \mu\text{m}^{-2}$
σ_{pp}	Particle diameter	$0.2\mu\text{m}$
v_0	Self-propulsion velocity	Varied systematically
N	Number of particles	60
Vesicle		
κ	Bending rigidity	$30k_B T$
D_m	Diffusion constant vertices	$0.1\mu\text{m}^2/\text{s}$
Δt_{bf}	Bond flip interval	$20\Delta t$
A_0	Center value area potential	$4\pi R_0^2$
M_A	Stiffness area potential	$10^6 k_B T$
V_0	Center value volume potential	$\frac{4}{3}\pi R_0^3$
M_V	Stiffness volume potential	$0, 10^6 k_B T$
R_0	Radius reference vesicle	$1\mu\text{m}$
N_v	Number of vertices membrane mesh	600
Global parameters		
k_{pv}	Stiffness particle-vertex interaction	$10^6 k_B T \mu\text{m}^{-2}$
σ_{pv}	Particle-vertex cross-section	$0.2\mu\text{m}$
Δt	Time step	10^{-6}s

TABLE S1. Baseline parameter values for the simulations of the vesicle containing the active particles (Figs. 1, 3 and 4 of the main text). For the phase diagram of the vesicle in which both the area and volume are constrained (Fig. 1a main text), $M_V = 10^6$. For the vesicles in which the volume can change freely (Figs. 1, 3, 4 main text and Fig. S2), $M_V = 0$. The parameters have been chosen such that the predictions of our work can be tested experimentally. Specifically, the bending rigidity corresponds to that of a typical Giant Unilamellar Vesicle [5], while the self-propulsion velocity, diffusion constants and particle size correspond to those of self-propelled colloids [6]. The parameters to model the interaction between the vertices were taken from Ref. [3].

The prolateness cannot be used, however, to reliably differentiate the stomatocytes from the prolate and oblate shapes, because for the stomatocytes $\mathbb{P}$ fluctuates around zero. To identify the stomatocytes we therefore employed the aspheriticity $\mathbb{A}$, which is defined as

$$\mathbb{A} = \frac{(\lambda_2 - \lambda_1)^2 + (\lambda_3 - \lambda_1)^2 + (\lambda_3 - \lambda_2)^2}{2(\lambda_1 + \lambda_2 + \lambda_3)^2}. \quad (\text{S17})$$

It characterizes how far a shape deviates from that of the ideal sphere. A vesicle was identified as a stomatocyte when $\mathbb{A} < 0.1$ and its reduced volume is $\hat{V} < 0.7$. The reduced volume criterion allows us to distinguish between fairly spherical vesicles with a large reduced volume and stomatocytes, which have a significantly smaller reduced volume.

SWIM PRESSURE COMPUTATION

The active particles are confined between two sinusoidal surfaces described by $z_w(x, y) = z_0 + B \sin(2\pi x/L) \sin(2\pi y/L)$ and $-z_w(x, y)$, respectively, where z_0 has been chosen large enough such that the correlation between the particle distributions at the two respective surfaces vanishes. Periodic boundary conditions

are employed in both the x and y directions, and for simplicity the periods are taken to be equal, $L_x = L_y = L$. Similar to Eqs. S1 and S2, the motion of the active particles is given by

$$\dot{\mathbf{r}}_i = \mu \mathbf{F}_p^{(i)}(\mathbf{r}_i) + v_0 \mathbf{e}_i, \quad (\text{S18})$$

$$\dot{\mathbf{e}}_i = \sqrt{2D_r} \boldsymbol{\chi}_r(t) \times \mathbf{e}_i, \quad (\text{S19})$$

where μ is the mobility of the active particles. Following Nikola *et al.* [9], we consider the limit in which the thermal translational diffusion constant of the particles vanishes, while the mobility of the particles μ remains finite. This allows us to zoom in on the effect of the active swim pressure on the total pressure. The parti-

Parameter	Description	Value
μ	Motility	$1\mu\text{m}^2\text{s}^{-1} k_B T^{-1}$
L	Period	$1\mu\text{m}$
λ	Stiffness wall potential	$10^3 k_B T \mu\text{m}^{-2}$
D_r	Rotational diffusion constant	$1/\text{s}$
$2z_0$	Separation between surfaces	$60\mu\text{m}$

TABLE S2. Simulation parameter values for the computation of the swim pressure as a function of the curvature, corresponding to Fig. 2 of the main text.

cles do not interact with each other yet do interact with the wall with a force $\mathbf{F}_p^{(i)} = -\nabla_i V_w$, where $V_w(\mathbf{r}_i) = (1/2)\lambda[z_i - z_w(x_i, y_i)]^2$ is the interaction potential between particle i with position $\mathbf{r}_i = (x_i, y_i, z_i)$, and the surface. In the simulations, the particles are propagated via an Euler scheme similar to Eqs. S12 and S13:

$$\Delta \mathbf{r}_i = \mu \mathbf{F}_p^{(i)}(\mathbf{r}_i) \Delta t + v_0 \mathbf{e}_i \Delta t, \quad (\text{S20})$$

$$\Delta \mathbf{e}_i = \sqrt{2D_r \Delta t} \chi_r(t) \times \mathbf{e}_i, \quad (\text{S21})$$

The local swim pressure per unit bulk concentration of the active particles $p_{\text{swim}}(x, y)$ is computed in the simulations from the average

$$p_{\text{swim}}(x, y) = \frac{\langle \partial_z V_w(\{\mathbf{r}_i\}) \rangle_{x,y}}{\rho_b}, \quad (\text{S22})$$

where ρ_b is the bulk concentration of the active particles.

For comparison, we also compute p_{swim} in a 2 dimensional system with $z_w(x) = z_0 + B \sin(2\pi x/L)$ as in Nikola *et al.* [9].

In the simulations, we vary H and C by varying B , keeping L constant. The parameter values are given in Tab. S2.

THE EFFECT OF EXCLUDED-VOLUME INTERACTIONS BETWEEN THE ACTIVE PARTICLES ON THE SWIM PRESSURE

Fig. 2 of the main text shows that in 3D the swim pressure increases superlinearly with the membrane curvature while in 2D it increases linearly or even weakly sublinearly. The principal reason for this behavior is that in 3D there is one more dimension in which the membrane is curved, which means that the particles accumulate more strongly with curvature in 3D than in 2D. To elucidate this interaction between membrane curvature and particle accumulation, and its dependence on dimensionality, the particles were modeled as ideal particles. Yet in our main model, underlying Figs. 1, 3 and 4 of the main text, the particles have a finite size, and thus exhibit excluded-volume interactions with one another. Here we ask whether these excluded-volume interactions change the super- versus (sub) linear dependence of the swim pressure on membrane curvature in 3D and 2D, respectively.

To this end we recomputed the swim pressure generated by finite-sized particles between two sinusoidal surfaces. The set up is identical to that of Fig. 2 of the main text, including the membrane-particle interaction, except that the particles have an excluded-volume interaction with one another as given by Eq. S3 (and as in the model corresponding to Figs. 1, 3, and 4 of the main text). In Eqs. S18 and S20, now the force on particle i is $\mathbf{F}_p^{(i)}(\{\mathbf{r}_i\}) = -\nabla_i U_p(\{\mathbf{r}_i\})$, where the total energy $U_p(\{\mathbf{r}_i\}) = V_w(\mathbf{r}_i) + U_{\text{pp}}(\{\mathbf{r}_i\})$ of the particles is the sum of the interaction energy with the

wall $V_w(\mathbf{r}_i) = (1/2)\lambda[z_i - z_w(x_i, y_i)]^2$ and the particle-particle interaction energy $U_{\text{pp}}(\{\mathbf{r}_i\})$ as given by Eq. S3. The volume fractions of the particles were chosen such that they are comparable to the volume fraction of the particles in our vesicle at the onset of the sphere-to-prolate transition.

Fig. S1 shows the result. Comparing the behavior of the finite-sized particles (red and blue dots) with that of the ideal particles (black dots), it is clear that the excluded volume interactions lower the absolute value of the swim pressure. This is because these interactions cause the particles to block each other, lowering the pressure they exert on the membrane [10]. However, also with excluded-volume interactions the swim pressure continues to scale superlinearly with membrane curvature in 3D and (sub) linearly with membrane curvature in 2D. Excluded-volume interactions thus do not change the scaling behavior of the swim pressure with membrane curvature, at least not for the volume fraction of the particles in our model.

ELASTIC PRESSURE

The *closed* membrane of the vesicle is simulated with an area constraint, see Eq. S7. However, deforming a *planar* membrane is impossible without a change of its area. To study the curvature dependence of the interplay between the elastic pressure and the active swim pressure as described in the previous section, we introduce a surface tension γ and add an energy term $\mathcal{H}_A = \gamma A$ to the Helfrich bending free energy $\mathcal{H}_{\text{bend}}$ to describe the energy of the surface: $\mathcal{H} = \mathcal{H}_{\text{bend}} + \mathcal{H}_A$. At the apex, the resulting elastic pressure is

$$\begin{aligned} \Delta p_{\text{elastic}}(r_{\text{apex}}) &= \gamma \nabla^2 h + \kappa \nabla^4 h = 2\gamma H + 2\kappa \nabla^2 H \\ &= 2\gamma H + 2q_0^2 \kappa H \end{aligned} \quad (\text{S23})$$

where h is the height of the membrane away from the reference plane and $q_0 = 2\pi/L$ is the lowest wave-number with periodic length L . Similarly, in the 2D semi-flexible polymer case $\Delta p_{\text{elastic}} = \gamma C + q_0^2 \kappa C$, where C is the curvature of the line. Clearly, at the apex $\Delta p_{\text{elastic}}$ varies linearly with the curvature both in 3D and 2D.

THE EFFECT OF THE TRANSLATIONAL DIFFUSION CONSTANT

Fig. S2 shows that when the translational diffusion constant D_t is increased, both the coexistence velocity v_0^{coex} and the barrier height increase. Increasing D_t tends to decrease the propulsion force $f_{\text{prop}} = v_0 k_B T / D_t$. To keep the system at coexistence, the coexistence propulsion velocity v_0^{coex} must increase as D_t is raised. Increasing v_0 increases the density of particles at the surface [11, 12], which tends to increase the total force on the membrane. To remain at coexistence, the force per particle, f_{prop} , must therefore decrease when v_0^{coex} increases.

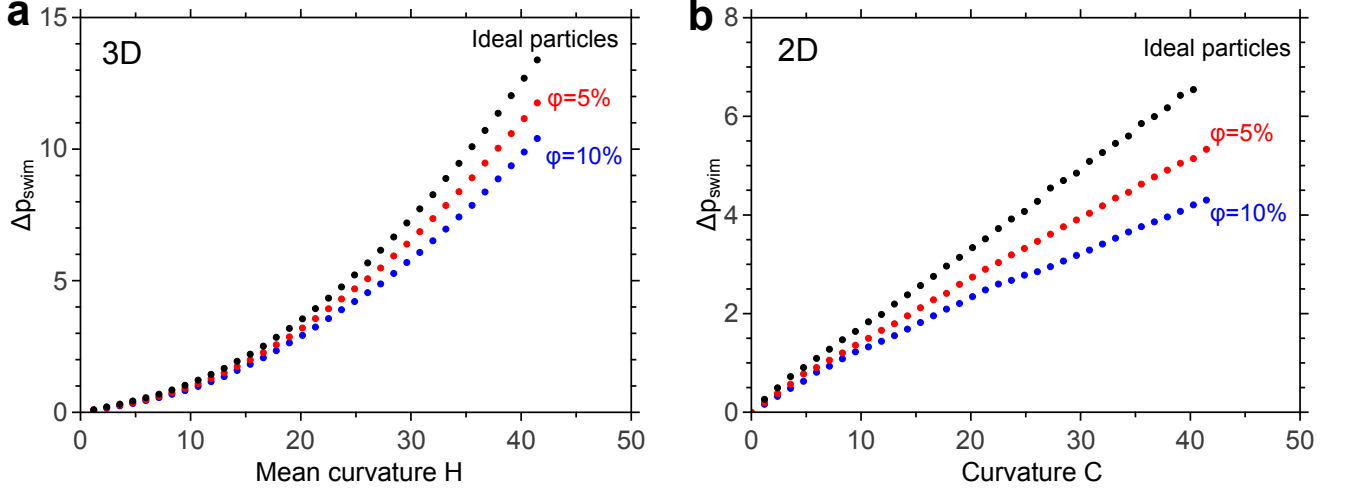


FIG. S1. **Excluded volume interactions between the active particles lower the swim pressure but do not change scaling of the swim pressure with membrane curvature.** **a** The swim-pressure difference Δp_{swim} for particles between the outer and inner apices of a sinusoidal surface in 3D as a function of the absolute local mean curvature H at the apices, both for ideal particles that do not interact with each other and for particles with excluded volume interactions, at two different volume fractions φ . It is seen that the excluded-volume interactions lower the swim pressure, yet the swim pressure continues to scale super-linearly with membrane curvature. **b** The swim-pressure difference between the outer and inner apices for a sinusoidal line in 2D as a function of the absolute curvature C , again for ideal particles and finite-sized particles at two different area fractions φ . The excluded volume interactions lower the swim pressure yet the swim pressure continues to scale (sub) linearly with membrane curvature. Following Fig. 2 of the main text, the swim pressure is per unit density of the active particles. Parameters: To get good statistics on the swim pressure computation we had to change the particle diameter from $\sigma = 0.2\mu\text{m}$ (corresponding to Figs. 1, 3, 4 of the main text) to $\sigma = 0.3\mu\text{m}$ and the surface separation in 3D from $2z_0 = 60\mu\text{m}$ (Fig. 2 of the main text) to $2z_0 = 10\mu\text{m}$; the other parameter values have not been changed and can be found in Table S2.

Interestingly, the simulations reveal that v_0^{coex} grows sub-linearly with D_t , which means that $f_{\text{prop}}^{\text{coex}} = v_0^{\text{coex}} k_B T / D_t$ indeed decreases. Lastly, because the force per particle $f_{\text{prop}}^{\text{coex}}$ decreases, more particles must participate in nucleating the shape transformation, making it more collective and raising the barrier.

FFS SIMULATIONS

We use Forward Flux Sampling (FFS) to enhance the sampling of the rare transition between the (nearly) spherical shape and the distinctly prolate shape when the volume of the vesicle is allowed to vary freely, see Fig. 1 of the main text [13]. To enhance the sampling, FFS employs a series of interfaces $\{\lambda_0, \lambda_1, \dots, \lambda_{n-1}, \lambda_n\}$ in between the initial state A and final state B . FFS then exploits that the rate constant for the transition from A to B can be written as [13, 14]

$$k_{AB} = \Phi_A \prod_{i=0}^{n-1} P(\lambda_{i+1} | \lambda_i). \quad (\text{S24})$$

Here, Φ_A is the flux of trajectories starting in A and crossing the first interface λ_0 , while $P(\lambda_{i+1} | \lambda_i)$ is the probability that a trajectory which crosses interface λ_i reaches the next interface λ_{i+1} before it returns to A .

These interfaces allow FFS to drive the system over the barrier in a ratchet-like manner: rather than waiting for the rare event in which the system makes a complete transition from A to B to happen, FFS splits up the problem by performing a series of short simulations that take the system from one interface to the next. Here, we use the “direct” variant of FFS, DFFS, in which all members of the transition path ensemble are grown in concert, rather than one at the time [15]. In essence, we first perform a brute-force simulation in the initial state, where each time the trajectory crosses the first interface λ_0 the configuration is stored. This brute-force simulation yields an estimate for the flux Φ_A and an ensemble of configurations at the first interface λ_0 . In the next step, we launch many trial trajectories from the configurations at λ_0 ; these trajectories are propagated until they either return to A or reach the next interface λ_1 , upon which the configuration is stored. The fraction of trajectories that makes it to λ_1 provides an estimate for $P(\lambda_1 | \lambda_0)$, while the list of configurations at λ_1 provides the starting point for a next series of simulations in which the system is driven from λ_1 to λ_2 . Indeed, we repeat this procedure for each interface, until we have computed all the required crossing probabilities $P(\lambda_{i+1} | \lambda_i)$. The order parameter that we use to define the interfaces is the reduced volume: $\lambda_i = \hat{V}_i$.

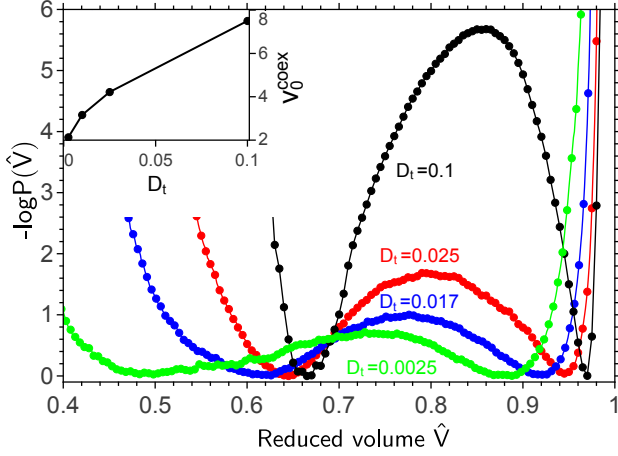


FIG. S2. **The self-propulsion velocity and barrier height at sphere-prolate coexistence depend on the translational diffusion constant of the particles.** Generalized free energy $-\log P(\hat{V})$ for different values of the particles' translational diffusion constant D_t , with in the inset v_0^{coex} as a function of D_t . Both the barrier height and v_0^{coex} increase with D_t . Other parameter values as in Fig. 1 of the main text (see Table S1); only the membrane area, and not the volume, is constrained.

To compute the stationary distribution $P(\hat{V})$ we perform a series of FFS simulation in the forward direction from A (prolate) to B (sphere) and another one in the reverse direction, from B to A [16]. These two FFS simulations make it possible to compute $P(\hat{V})$ via

$$P(\hat{V}) = \psi_A(\hat{V}) + \psi_B(\hat{V}). \quad (\text{S25})$$

Here $\psi_A(\hat{V})$ is the contribution to $P(\hat{V})$ from the paths that start in A and either make it to B or return to A , while $\psi_B(\hat{V})$ is the contribution from paths in the opposite direction, starting in B and ending in either A or B . These contributions are given by

$$\psi_A(\hat{V}) = p_A \Phi_A \tau_+(\hat{V}; \lambda_0) \quad (\text{S26})$$

$$\psi_B(\hat{V}) = p_B \Phi_B \tau_-(\hat{V}; \lambda_n) \quad (\text{S27})$$

Here p_A the steady-state probability of finding the system in A , Φ_A is defined as in Eq. S24, and $\tau_+(\hat{V}; \lambda_0)$ is the average time a trajectory, which originates at λ_0 , spends at value $\hat{V}$. The quantities for ψ_B are defined analogously. The quantities p_A and p_B can be computed once the rate constants k_{AB} and k_{BA} have been computed with FFS, using $p_A k_{AB} = p_B k_{BA}$. The function $\tau_+(\hat{V}; \lambda_0)$ is given by

$$\tau_+(\hat{V}; \lambda_0) = \pi_+(\hat{V}; \lambda_0) + \sum_{i=1}^{n_+-1} \pi_+(\hat{V}; \lambda_0) \prod_{j=0}^{i-1} P(\lambda_{j+1} | \lambda_j), \quad (\text{S28})$$

i	$\hat{V}_i^+$	$\hat{V}_i^-$
0	0.71	0.95
1	0.712	0.949
2	0.715	0.948
3	0.72	0.946
4	0.735	0.943
5	0.76	0.934
6	0.79	0.915
7	0.82	0.89
8	0.85	0.86
9	0.88	0.81
10	0.92	0.76
11	0.95	0.71

TABLE S3. Positions of the interfaces $\lambda_i = \hat{V}_i$ used for the FFS simulations of Fig. 1 of the main text, for $v_0 = 7.5 \mu\text{m/s}$. The positions for the transition in the forward direction from the prolate shape (state A) to the nearly spherical shape (state B) are denoted by $\hat{V}_i^+$ and those for the reverse transition by $\hat{V}_i^-$. The positions of the interfaces in the other FFS simulations are slightly different, to optimize the spacing between them.

where $\pi_+(\hat{V}; \lambda_0)$ denotes the average time a trajectory coming from interface λ_0 spends at order parameter value $\hat{V}$. The function $\tau_-(\hat{V}; \lambda_n)$ in $\psi_B(\hat{V})$ is defined similarly.

Since the optimal spacing between the interfaces is smaller at the pre-critical side of the barrier than at the post-critical side, we positioned the interfaces differently for the forward and reverse transitions. The positions for the FFS simulations of Fig. 1 of the main text corresponding to $v_0 = 7.5 \mu\text{m/s}$ are given in Table S3. For the other FFS simulations slightly different positions for the interfaces were used to optimize the spacing between them. The number of configurations at each interface is typically 1000–10000 while the number of trial runs is on the order of 3000–30000. The length of a typical transition path is 4×10^6 time-steps, corresponding to a duration of 4s. For the baseline parameters given in Table S1, the rate constant $k_{AB} = k_{BA} = k$ for the prolate $\leftrightarrow$ sphere transition at coexistence, $v_0 = 7.5 \mu\text{m/s}$ (see Fig. 1 main text), is $k = 5 \times 10^{-4} \text{s}^{-1}$.

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